

LMI-based Formation Control with Collision Avoidance for Air-Ground Fleet

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Abstract: This paper proposes a formation controller for a fleet composed of two unicycles and a quadcopter. The formation problem is defined as a min-max problem whose optimal control strategy is linear in the local measurements of each agent, and the matrix gains are obtained by solving a Linear Matrix Inequality. Artificial repulsive forces are added to avoid inter-agent collisions and obstacles. The proposed control scheme was validated both in simulation and through several experiments, involving both static and dynamic obstacles. The results show that the agents can achieve the formation without crashes.

Keywords: Nonlinear and optimal automotive control, Adaptive and robust control of automotive systems, Vehicle dynamic systems, Cooperative Nonlinear Control, Robust Linear Matrix Inequalities.

1. INTRODUCTION

Multi-agent systems (MASs) are nowadays used in a variety of applications as they can perform complex tasks, which would be otherwise impossible for a single robot. This is evident when considering an heterogeneous fleet where the strengths of one type of robot can compensate for the limitations of another. For example, to monitor a large area, multiple ground agents can move from one point to another while being supervised from above by a drone with a wide field of view. A deeper analysis of the subject to understand the potential tasks and challenges of MASs, the reader is referred to the work of ?.

One of the fundamental cooperative tasks when working with multi-agents is formation control. This topic has been studied for years. For example, ? already proposed a decentralized control algorithm for a team of two-wheeled robots to achieve a geometric pattern. However, formation control remains an active research topic. ? experimentally validated a robust distributed control based on negative imaginary systems consensus theory, using both ground robots and an air-ground fleet. In the work of ?, each agent computes its control action in a leader-follower manner using local extended Kalman filter's estimates to achieve a desired shape together with other wheeled robots and a drone. An optimal distributed formation control algorithm for double-integrator multi-agents has been presented and validated through simulations by ?. ? defined the formation problem for a group of double-integrator agents as a discrete-time game whose solution is given by a state-dependent Riccati equation. A robust distributed consensus controller is presented by ? to address the rendezvous

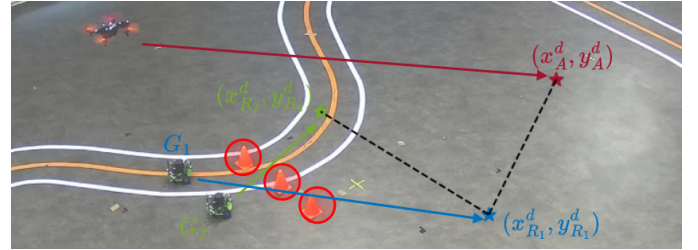


Fig. 1. Heterogeneous multi-agent setup for the experimental validation of the proposed formation control, including collision and obstacle avoidance.

problem and applied in simulation to a swarm of drones. In the recent years, reinforcement learning has been used in many areas including formation control. ? addressed multi-particle formation control by combining graph attention networks and multiple long short-term memories to achieve the desired shape and avoid collisions respectively. A position and an orientation robust controllers based on reinforcement learning are proposed and validated through simulations by ? for the formation of a team of quadrotors.

This is the context of our work, which proposes a novel formation control scheme for a group consisting of a quadcopter and two unicycles, as shown in Fig. ???. When addressing the multi-agent formation, there are two main challenges to solve: achieving the desired shape and avoiding collisions. On one hand, for the first challenge, a distributed controller combining Feedback Linearization (FL) and an optimal linear controller based on Linear Matrix Inequalities (LMIs) is presented. LMIs are a powerful mathematical tool used in many applications, among

which is also formation control. In the work of ?, a LMI is solved to design the controller and observer gains, guaranteeing stable formation flight for a group of quadcopters. ? propose a formation control with artificial fixed delays for multiple double-integrators and verify the asymptotic stability of the scheme by checking the solvability of a LMI. ? present a LMI formulation of the LQR problem to guarantee stable state consensus with an optimal control effort in a multi-agent system. On the other end, to avoid collision avoidance, the chosen solution is Artificial Potential Field (APF). In analogy with a particle moving in an electrostatic or gravitational field, virtual repulsive force fields are simulated to ward off the robot from an obstacle. This approach has been widely used in formation control of ground vehicles, see for example the work of ?. However, in more recent works, e.g. ? and ?, APFs are employed to navigate around obstacles and prevent collisions while attaining the desired shape.

This paper represents the continuation of the previous work by ?, where a control architecture combining Feed-back Linearization (FL), a robust LMI-based controller, and APFs was proposed and validated through simulations for the formation of three unicycles. The successive developments sought to manage the challenge of a heterogeneous team (replacing a ground vehicle with a quadcopter) by adapting the methodology and then experimentally validating the new solution. The contributions of this work are: a) a controller involving FL, a LMI-based controller and APFs is presented for the stable formation of an air-ground fleet; b) the proposed controller has been validated through MATLAB Simulink simulations and a comprehensive set of experiments in an indoor arena, including both static and dynamic obstacle scenarios. The remainder paper is structured as follows. In Section ??, the mathematical model used to describe the team is stated. The proposed formation controller is detailed in Section ?. The simulations and the experiments results are reported and analyzed in Section ?? and in Section ?? respectively. Finally, the conclusions are presented in Section ?.

2. HETEROGENEOUS FLEET MODELING

In this section, the dynamics of the agents composing the team (i.e., of the two unicycles and the quadrotor, as shown in Fig. ??) are described and adapted to then apply the LMI approach. The well-known kinematic model of a two-wheeled ground robot G_i is

$$\begin{cases} \dot{x}_i &= v_i \cos \theta_i \\ \dot{y}_i &= v_i \sin \theta_i \\ \dot{\theta}_i &= \omega_i \end{cases} \quad (1)$$

where (x_i, y_i, θ_i) is the pose of the agent and (v_i, ω_i) are the linear and angular velocities respectively. The equations (??) describing the motion of the center of rotation are non-linear. Still, if a point located at distance $R > 0$ from (x_i, y_i) , which will be denoted as (x_{Ri}, y_{Ri}) , is considered and new virtual inputs are defined $(u_{i,x}, u_{i,y})$, the dynamics become linear:

$$\begin{cases} \dot{x}_{Ri} &= u_{i,x} \\ \dot{y}_{Ri} &= u_{i,y} \end{cases} \quad (2)$$

Moving on to the drone, since the formation problem is restricted to the (X, Y) -plane, some simplifying hypothesis

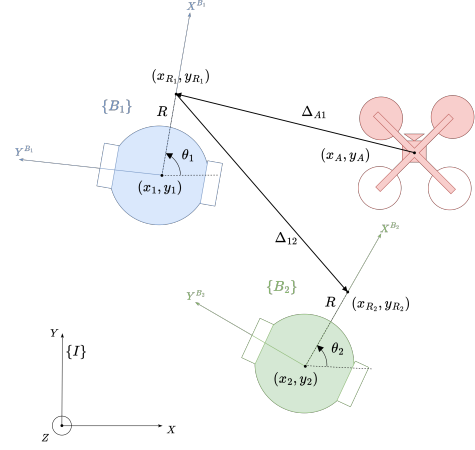


Fig. 2. Air-ground heterogeneous fleet.

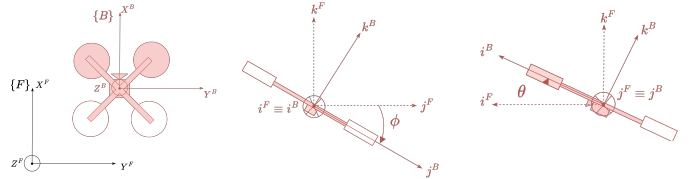


Fig. 3. Quadcopter's roll ϕ and pitch θ angles.

have been made. First, let us assume that there is an inner attitude controller ensuring attitude stability. Hence, the focus is only on the translational dynamics which are:

$$\begin{cases} m\ddot{x}_A^F &= -T \sin \theta \\ m\ddot{y}_A^F &= T \cos \theta \sin \phi \\ m\ddot{z}_A^F &= T \cos \theta \cos \phi - mg \end{cases} \quad (3)$$

Let us analyze the equations with the help of Fig. ?. The state (x_A^F, y_A^F, z_A^F) represents the position of the drone w.r.t. the inertial frame $\{F\}$. The frame $\{F\}$ has been used instead of $\{I\}$ to be aligned with the framework employed to implement the controller for the drone, see Section ?. The angles ϕ and θ are the roll and pitch Euler angles of the body frame $\{B\}$ (fixed to the drone) relative to the inertial frame $\{F\}$. Ignoring the small body forces, the only forces acting on the drone are the thrust produced by the motors T along the k^B direction and the gravity mg along the $-k^F$ direction. Let us suppose that the Z -controller

$$T = (r_1 + mg)/(\cos \phi \cos \theta) \quad (4)$$

is such that the drone's mass is compensated in a short time by r_1 , i.e. $r_1 \rightarrow 0$. This implies that the (x, y) -dynamics will not include the mass of the drone, as the Z -controller already compensated it. Finally, let us make the hypothesis that the roll and pitch angles are small, i.e. $\phi, \theta \approx 0$. Then, the translational dynamics simplify to

$$\begin{cases} \ddot{x}_A^F &= -g\theta \\ \ddot{y}_A^F &= g\phi \end{cases} \quad (5)$$

At this point, all the agents can be described by the linear models (??) and (??). To define the formation problem, let us introduce the quantities

$$\Delta_{A1} = [\Delta_{A1,x} \quad \Delta_{A1,y}]^T \quad (6a)$$

$$\Delta_{A1,x} = x_{R1} - x_A = x_{R1} - y_A^F \quad (6b)$$

$$\Delta_{A1,y} = y_{R1} - y_A = y_{R1} - x_A^F \quad (6c)$$

with (x_A, y_A) the position of the drone w.r.t. the inertial frame $\{I\}$,

$$\Delta_{12} = [\Delta_{12,x} \ \Delta_{12,y}]^T \quad (7a)$$

$$\Delta_{12,x} = x_{R_2} - x_{R_1}, \ \Delta_{12,y} = y_{R_2} - y_{R_1} \quad (7b)$$

the desired inter-distances Δ_{A1}^d and Δ_{12}^d , and the desired position of the drone (x_A^d, y_A^d) w.r.t. the inertial frame $\{I\}$ (constant over time). The overall dynamics of the formation's errors are as follows

$$\begin{cases} \ddot{e}_{A,x} = -g\theta \\ \ddot{e}_{A,y} = g\phi \\ \dot{e}_{\Delta_{A1},x} = u_{1,x} - \dot{e}_{A,y} \\ \dot{e}_{\Delta_{A1},y} = u_{1,y} - \dot{e}_{A,x} \\ \dot{e}_{\Delta_{12},x} = u_{2,x} - u_{1,x} \\ \dot{e}_{\Delta_{12},y} = u_{2,y} - u_{1,y} \end{cases} \quad (8)$$

with

$$e_{A,x} = x_A^F - y_A^d, \ e_{A,y} = y_A^F - x_A^d \quad (9)$$

$$e_{\Delta_{A1},x} = \Delta_{A1,x} - \Delta_{A1,x}^d, \ e_{\Delta_{A1},y} = \Delta_{A1,y} - \Delta_{A1,y}^d \quad (10)$$

$$e_{\Delta_{12},x} = \Delta_{12,x} - \Delta_{12,x}^d, \ e_{\Delta_{12},y} = \Delta_{12,y} - \Delta_{12,y}^d \quad (11)$$

Let us define the state and control vectors as follows

$$\mathbf{e}^T = [e_{A,x} \ \dot{e}_{A,x} \ e_{A,y} \ \dot{e}_{A,y} \ e_{\Delta_{A1},x} \ e_{\Delta_{A1},y} \ e_{\Delta_{12},x} \ e_{\Delta_{12},y}] \quad (12a)$$

$$\mathbf{u}_A = \begin{bmatrix} \phi \\ \theta \end{bmatrix}, \ \mathbf{u}_1 = \begin{bmatrix} u_{1,x} \\ u_{1,y} \end{bmatrix}, \ \mathbf{u}_2 = \begin{bmatrix} u_{2,x} \\ u_{2,y} \end{bmatrix}, \ \mathbf{u} = \begin{bmatrix} \mathbf{u}_A \\ \mathbf{u}_1 \\ \mathbf{u}_2 \end{bmatrix} \quad (12b)$$

Hence, the heterogeneous fleet's dynamics can be written in a matrix form:

$$\dot{\mathbf{e}} = \mathbf{A}\mathbf{e} + \mathbf{B}\mathbf{u} \quad (13)$$

By discretizing with sampling time ΔT using the Euler method:

$$\mathbf{e}(k+1) = \mathbf{A}_d \mathbf{e}(k) + \mathbf{B}_d \mathbf{u}(k) \quad (14)$$

with $\mathbf{A}_d = \mathbf{I} + \Delta T \mathbf{A}$ and $\mathbf{B}_d = \Delta T \mathbf{B}$. To model real-world disturbances, it is assumed that an additive zero-mean disturbance $\mathbf{d}(k)$ is acting on the input channel, i.e.

$$\mathbf{e}(k+1) = \mathbf{A}_d \mathbf{e}(k) + \mathbf{B}_d [\mathbf{u}(k) + \mathbf{d}(k)] \quad (15)$$

Let us make the following hypothesis: a) the drone knows its position; b) the robot G_1 knows its position w.r.t. the quadcopter; c) the robot G_2 knows its position w.r.t. the other agent G_1 ; d) all unicycles are aware of their distance from any obstacles on the ground; e) a hierarchy is defined between the ground robots: G_1 has the priority over G_2 .

3. FORMATION CONTROLLER

This section aims to detail the proposed formation controller. The two crucial issues, formation maintenance and collision avoidance, are addressed respectively in the next two subsections.

3.1 Formation Maintenance: a LMI Approach

The first goal is to ensure the robots achieve the desired shape while using only local measurements and counter-acting the worst-case disturbance. Based on the previous hypotheses, the measurements of each agent are as follows:

$$\mathbf{z}_A = \begin{bmatrix} e_{A,x} \\ e_{A,y} \\ \dot{e}_{A,x} \end{bmatrix} = \mathbf{C}_A \mathbf{e} \quad (16a)$$

$$\mathbf{z}_1 = \begin{bmatrix} e_{\Delta_{A1},x} \\ e_{\Delta_{A1},y} \end{bmatrix} = \mathbf{C}_1 \mathbf{e}, \ \mathbf{z}_2 = \begin{bmatrix} e_{\Delta_{12},x} \\ e_{\Delta_{12},y} \end{bmatrix} = \mathbf{C}_2 \mathbf{e} \quad (16b)$$

with $\mathbf{C}_A$, $\mathbf{C}_1$, and $\mathbf{C}_2$ the measurement matrices of the three. The idea is to formulate the robust formation control problem as a dynamic game between two players: one is the control action and the other opponent is the "nature". In the proposed solution, at each time step k the control action to be applied at the next step $\mathbf{u}(k)$ is obtained by solving the following min-max problem defined over a one-step horizon.

$$\begin{aligned} \min_{\mathbf{u}(k)} \max_{\mathbf{e}(k) \neq 0} \quad & \frac{J(k)}{\|\mathbf{e}(k)\|_2^2} \\ \text{s.t.} \quad & \mathbf{z}_i(k) = \mathbf{C}_i \mathbf{e}(k) \\ & \mathbf{u}_i(k) = \boldsymbol{\mu}_i(\mathbf{z}_i(k)) \quad i \in \{A, 1, 2\} \end{aligned} \quad (17)$$

with

$$J(k) = \mathbf{u}(k)^T \mathbf{Q}_u \mathbf{u}(k) + \mathbf{e}(k+1)^T \mathbf{Q}_e \mathbf{e}(k+1) + \mathbf{e}(k)^T \mathbf{Q}_e \mathbf{e}(k) \quad (18)$$

The cost function can be rewritten by substituting the evolution model equation (??).

$$J(k) = \begin{bmatrix} \mathbf{e}(k) \\ \mathbf{u}(k) \end{bmatrix}^T \begin{bmatrix} \mathbf{Q}_e + \mathbf{A}_d^T \mathbf{Q}_e \mathbf{A}_d & \mathbf{A}_d^T \mathbf{Q}_e \mathbf{B}_d \\ \mathbf{B}_d^T \mathbf{Q}_e \mathbf{A}_d & \mathbf{Q}_u + \mathbf{B}_d^T \mathbf{Q}_e \mathbf{B}_d \end{bmatrix} \begin{bmatrix} \mathbf{e}(k) \\ \mathbf{u}(k) \end{bmatrix} \quad (19)$$

Hence, the min-max problem to be solved is

$$\begin{aligned} \min_{\mathbf{u}(k)} \max_{\mathbf{e}(k) \neq 0} \quad & \frac{\begin{bmatrix} \mathbf{e}(k) \\ \mathbf{u}(k) \end{bmatrix}^T \begin{bmatrix} \mathbf{Q}_{ee} & \mathbf{Q}_{eu} \\ \mathbf{Q}_{ue} & \mathbf{Q}_{uu} \end{bmatrix} \begin{bmatrix} \mathbf{e}(k) \\ \mathbf{u}(k) \end{bmatrix}}{\|\mathbf{e}(k)\|_2^2} \\ \text{s.t.} \quad & \mathbf{z}_i(k) = \mathbf{C}_i \mathbf{e}(k) \\ & \mathbf{u}_i(k) = \boldsymbol{\mu}_i(\mathbf{z}_i(k)) \quad i \in \{A, 1, 2\} \end{aligned} \quad (20)$$

with

$$\mathbf{Q}_{ee} = \mathbf{Q}_e + \mathbf{A}_d^T \mathbf{Q}_e \mathbf{A}_d \quad (21)$$

$$\mathbf{Q}_{eu} = \mathbf{A}_d^T \mathbf{Q}_e \mathbf{B}_d, \ \mathbf{Q}_{ue} = \mathbf{Q}_{eu}^T \quad (22)$$

$$\mathbf{Q}_{uu} = \mathbf{Q}_u + \mathbf{B}_d^T \mathbf{Q}_e \mathbf{B}_d \quad (23)$$

Note that the problem (??) has the same form as the one considered by ? in their paper where they prove that the linear policy is optimal, i.e.

$$\mathbf{u}_i^*(k) = \mathbf{K}_i^* \mathbf{z}_i(k) \quad i \in \{A, 1, 2\} \quad (24)$$

where the matrices $\mathbf{K}_i^*$ can be obtained by solving the following LMI:

$$\begin{aligned} \min_{\gamma, \mathbf{K}} \quad & \gamma \\ \text{s.t.} \quad & \mathbf{K} = \text{diag}(\mathbf{K}_A, \mathbf{K}_1, \mathbf{K}_2) \\ & \begin{pmatrix} \mathbf{Q}_{ee} - \gamma \mathbf{I} + \mathbf{Q}_{eu} \mathbf{K} \mathbf{C} + \mathbf{C}^T \mathbf{K}^T \mathbf{Q}_{eu}^T & \mathbf{C}^T \mathbf{K}^T \\ \mathbf{K} \mathbf{C} & -\mathbf{Q}_{uu}^{-1} \end{pmatrix} \leq 0 \end{aligned} \quad (25)$$

where $\mathbf{C}$ is the matrix obtained by stacking the single $\mathbf{C}_i$. To conclude, please note that since the discrete-time system is time-invariant, the min-max problem (??) to be solved will be the same at each iteration. Hence, it is sufficient to solve the LMI (??) offline one time to obtain the optimal gain matrices $\mathbf{K}_i^*$, and then multiply at each instant k the local measurement $\mathbf{z}_i(k)$ by $\mathbf{K}_i^*$ to obtain $\mathbf{u}_i^*(k)$.

3.2 Collision and Obstacle Avoidance: APFs

Collisions can still occur when the agents attain the desired shape. Moreover, obstacles can be present in the robots' way. It is therefore crucial to propose a solution to avoid crashes for the experimental validation. This is where APFs come to help. For simplicity, let us consider at first a scenario where the ground is clean and collisions may occur

between G_1 and G_2 . On one hand, since G_1 has the highest priority, there is no need to correct the control action $\mathbf{u}_1^*(k)$. On the other hand, the robot G_2 has to adjust its way to avoid the other ground vehicle. A repulsive force can be used to attain this, with a direction opposite to the vector going from G_2 to G_1 and a magnitude as follows:

$$|F_{12}| = \begin{cases} (k_{Rep}/d_{Rep}^2) [(1/d_{12}) - (1/d_{Rep})] & d_{12} < d_{Rep} \\ 0 & d_{12} \geq d_{Rep} \end{cases} \quad (26)$$

with $d_{12} = \sqrt{\Delta_{12,x}^2 + \Delta_{12,y}^2}$ and d_{Rep} the distance after which the robot is not influenced by the obstacle. The final control inputs for G_2 are:

$$\begin{aligned} u_{2,x}(k) &= u_{2,x}^*(k) + \frac{1}{2} \frac{F_{12,x}(k)}{M} \Delta T \\ u_{2,y}(k) &= u_{2,y}^*(k) + \frac{1}{2} \frac{F_{12,y}(k)}{M} \Delta T \end{aligned} \quad (27)$$

with M the two-wheeled robots mass. The same approach can be extended to a scenario with both static and dynamic obstacles. Suppose that there are N (static or dynamic) obstacles, then the final control inputs for both unicycles are:

$$\begin{aligned} u_{1,x}(k) &= u_{1,x}^*(k) + \frac{1}{2} \frac{\sum_{j=1}^N F_{j1,x}(k)}{M} \Delta T \\ u_{1,y}(k) &= u_{1,y}^*(k) + \frac{1}{2} \frac{\sum_{j=1}^N F_{j1,y}(k)}{M} \Delta T \\ u_{2,x}(k) &= u_{2,x}^*(k) + \frac{1}{2} \frac{F_{12,x}(k) + \sum_{j=1}^N F_{j2,x}(k)}{M} \Delta T \\ u_{2,y}(k) &= u_{2,y}^*(k) + \frac{1}{2} \frac{F_{12,y}(k) + \sum_{j=1}^N F_{j2,y}(k)}{M} \Delta T \end{aligned} \quad (28)$$

4. SIMULATION RESULTS

The first validation of the proposed formation controller has been done via simulations in MATLAB R2022a Simulink. The tests have been carried out on a laptop with OS Ubuntu 20.04.6 LTS, equipped with 32 GB of RAM, an SSD NVMe, and a processor Intel Core i7-12800H. To choose the parameters, the target robots' characteristics have been taken into account from the beginning in the tuning phase. Two Waveshare JetBot Pro ROS AI and a Parrot AR.Drone 2.0 compose the experimental equipment specifically, see Section ???. First, both the linear and the angular velocities of the ground vehicles have been limited: $|v_i| \leq 0.5 \text{ m s}^{-1}$ and $|\omega_i| \leq 0.5 \text{ rad s}^{-1}$. Concerning the distance of the point from the center of rotation, it has been chosen $R = 0.18 \text{ m}$. The continuous-time models have been discretized with $\Delta T = 0.02 \text{ s}$. The weighting matrices used in the definition of the LMI have been set to $Q_e = 2e4 I_{8 \times 8}$ and $Q_u = \text{blkdiag}(0.01 I_{2 \times 2}, 0.04 I_{4 \times 4})$. The obtained K_i^* were

$$K_A^* = \begin{bmatrix} 0 & -0.0770 & 0 & -1.2440 \\ 0.0770 & 0 & 1.2440 & 0 \end{bmatrix} \quad (29)$$

$$K_1^* = \begin{bmatrix} -9.6717 & 0 \\ 0 & -9.6717 \end{bmatrix} \quad (30)$$

$$K_2^* = \begin{bmatrix} -15.0765 & 0 \\ 0 & -15.0765 \end{bmatrix} \quad (31)$$

which implies in closed loop the following eigenvalues of the closed-loop system

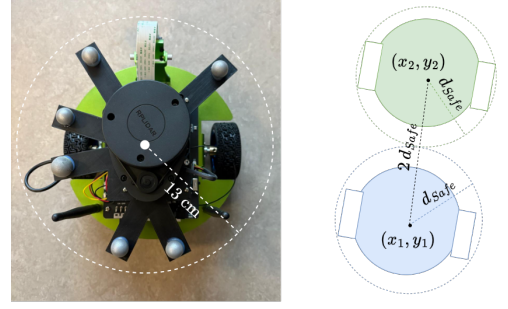


Fig. 4. The physical robot is contained entirely in a circle of radius $d_{Safe} = 0.13 \text{ m}$. Consequently, if the centers of rotation are at a distance less than $2d_{Safe}$ it means that the agents collided.

$$p = \text{eig}(A_d + B_d K^* C) = \begin{cases} 0.6985 & \mu = 2 \\ 0.7572 & \mu = 2 \\ 0.8066 & \mu = 2 \\ 0.9988 & \mu = 2 \end{cases} \quad (32)$$

which are all inside the unitary circle. Moving to the APFs, the mass of the Jetbot's robot is $M = 1.4 \text{ kg}$ and it has been chosen $k_{Rep} = 700$ and $d_{Rep} = 5R$. In the simulations, the unicycles are treated as mass points but nevertheless the objective is to verify if the physical robots will crash. For this reason, the robots are considered to have collided in the simulation if $d_{12} \leq 2d_{Safe}$ (with $d_{Safe} = 0.13 \text{ m}$, see Fig. ??). For a quantitative analysis, the time required for the agents to reach the formation was studied. Let us define the key performance index $\bar{t}$ as follows:

$$\forall t \geq \bar{t} : \begin{cases} |e_{A,x}(t)| < \varepsilon, |e_{A,y}(t)| < \varepsilon \\ |e_{\Delta_{A1},x}(t)| < \varepsilon, |e_{\Delta_{A1},y}(t)| < \varepsilon \\ |e_{\Delta_{12},x}(t)| < \varepsilon, |e_{\Delta_{12},y}(t)| < \varepsilon \end{cases} \quad (33)$$

with ε being a negligible error. For the simulations, it has been set to $\varepsilon = 0.10 \text{ m}$.

The first set of simulations considers a scenario with the ground clean and without disturbances, i.e. $\mathbf{d}(k) = 0$. The results are shown in Fig. ??. For each unicycle, the 2D-trajectories of the center of rotation are plotted along with the point (x_{R_i}, y_{R_i}) with a solid and a dotted line, respectively. The desired inter distances drone- G_1 and G_1 - G_2 are plotted in dotted black, while the actual inter distances at the end of the simulation are shown as solid black lines. Finally, stars denote the desired final positions of the robots based on the definitions of (x_A^d, y_A^d) , Δ_{A1}^d and Δ_{12}^d . From Fig. ??, it is straightforward to conclude that the agents make the formation avoiding collisions in all scenarios, which differ in initial conditions and desired geometric shapes. The key performance index for the first scenario is $\bar{t} = 59.38 \text{ s}$ and for the second case $\bar{t} = 59.38 \text{ s}$.

To demonstrate the importance of APFs, the simulations have been redone without repulsive forces. If a collision occurs this time, the ground vehicles will halt and remain stopped until the end of the test. Fig. ?? displays the resulting trajectories. Let us compare Fig. ?? and Fig. ?? with Fig. ?? and Fig. ??, respectively. Without APFs, the unicycles get too close and they must stop. Hence, even if the drone reaches the desired position (x_A^d, y_A^d) , the overall formation is not attained by the fleet. On the

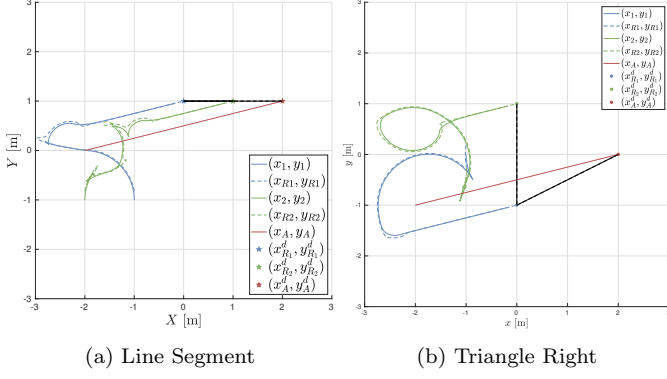


Fig. 5. Trajectories of the three agents with desired and actual interdistances in ideal simulations.

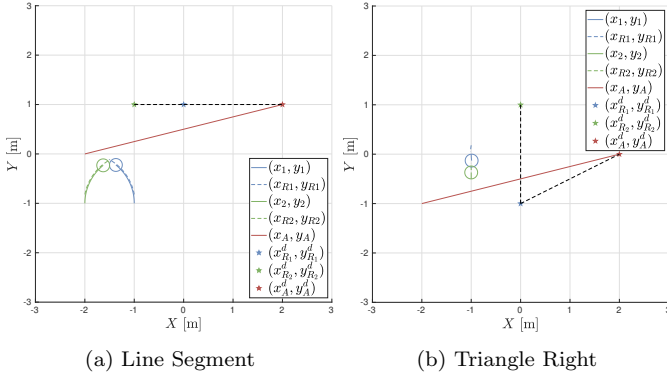


Fig. 6. Trajectories of the three agents without APFs in ideal simulations.

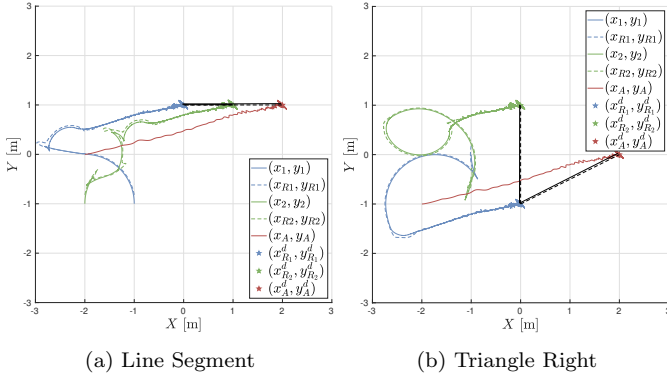


Fig. 7. Trajectories of the three agents with desired and actual interdistances in simulation with disturbances.

contrary, thanks to the artificial repulsive forces the robot G_2 deviates its path to stay at a safe distance from G_1 .

The last set of simulations aims to study the robustness of the proposed control scheme in the presence of the zero-mean disturbance $\mathbf{d}(k)$. The disturbance has been modeled in particular as a gaussian noise with variance $\Sigma = 0.1I_{6 \times 6}$. The results are presented in Fig. ???. The trajectories are similar to the ones in Fig. ??, but the agents in the end oscillate around the desired final positions due to disturbance $\mathbf{d}(k)$ (though they remain within a small neighborhood). This is the reason why the solid and dotted lines do not exactly overlap: the solid shape appears indeed to be a translated version of the dotted one. However, collisions between G_1 and G_2 are avoided and

the formation is reached after $\bar{t} = 112.14$ s and $\bar{t} = 112.1$ s, respectively.

5. EXPERIMENTAL VALIDATION

Once analyzed the simulations and given the promising results, the next step was to validate the formation control scheme through experiments. The tests have been carried out in an indoor arena equipped with a motion capture Optitrack system, used to measure the robots' positions. For the communication between multi-agents, the TCP protocol and ROS1 Noetic facilities have been used. The control actions were computed into separate ROS nodes running on a ground station (i.e., the same laptop used for the simulations) then sending the commands to the separate robots. For controlling the quadcopter, the implementation of the inner controller has been done using the FL-AIR framework developed by ?.

Initially, some preliminary tests were run using the gains used in the simulations, but the drone presented unwanted oscillations. Making an analogy with a PD controller, we would indeed like the gain K_A^* to have the following shape:

$$\begin{bmatrix} 0 & -K_P & 0 & -K_D \\ K_P & 0 & K_D & 0 \end{bmatrix} \quad (34)$$

Which is equivalent to two PD controllers, one for the X -dynamics and one for the Y -dynamics. Moreover, when tuning experimentally a PD attitude controller for the quadcopter, good performance were obtained with $K_D \approx 1.222K_P = \alpha K_P$. As tuning the weighting matrices to achieve the structure in (??) isn't trivial with the current definition of $\mathbf{z}_A$ in (??), the drone's measurement has been redefined to enforce this form, as follows:

$$\mathbf{z}'_A = \begin{bmatrix} e_{A,x} + \alpha \dot{e}_{A,x} \\ -e_{A,y} - \alpha \dot{e}_{A,y} \end{bmatrix} = C'_A e \quad (35)$$

The min-max problem (??) has been consequently changed to:

$$\begin{aligned} \min_{\mathbf{u}(k)} \max_{\mathbf{e}(k) \neq 0} & \frac{J(k)}{\|\mathbf{e}(k)\|_2^2} \\ \text{s.t. } & \mathbf{z}'_A(k) = C'_A \mathbf{e}(k) \\ & \mathbf{z}_i(k) = C_i \mathbf{e}(k) \quad i \in \{1, 2\} \\ & \mathbf{u}_A(k) = \boldsymbol{\mu}_A(\mathbf{z}'_A(k)) \\ & \mathbf{u}_i(k) = \boldsymbol{\mu}_i(\mathbf{z}_i(k)) \quad i \in \{1, 2\} \end{aligned} \quad (36)$$

With this new formulation, the weighting matrices have been chosen, following a tuning phase, as:

$$Q_e = \text{diag}(971.10, 971.10, 1000, 1000, 35, 35, 40, 40) \quad (37a)$$

$$Q_u = \text{diag}(0.01, 0.01, 0.08, 0.08, 0.08, 0.08) \quad (37b)$$

The resulting optimal gain matrices are

$$K_A'^* = \begin{bmatrix} -0.1258 & 0 \\ 0 & 0.1258 \end{bmatrix} \quad (38)$$

$$K_1^* = \begin{bmatrix} -7.8261 & 0 \\ 0 & -7.8261 \end{bmatrix} \quad (39)$$

$$K_2^* = \begin{bmatrix} -8.2934 & 0 \\ 0 & -8.2934 \end{bmatrix} \quad (40)$$

while the eigenvalues in closed-loop are:

$$p' = \text{eig}(A_d + B_d K'^* C') = \begin{cases} 0.8341 & \mu = 2 \\ 0.8435 & \mu = 2 \\ 0.9849 + 0.0163i & \mu = 2 \\ 0.9849 - 0.0163i & \mu = 2 \end{cases} \quad (41)$$

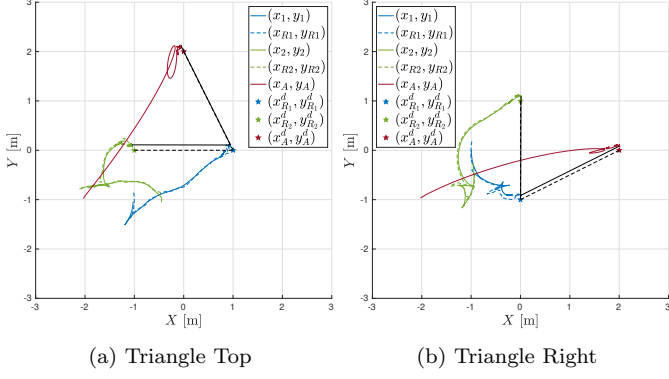


Fig. 8. Trajectories of the three agents with desired and actual interdistances in the experiments with the ground clear of obstacles.

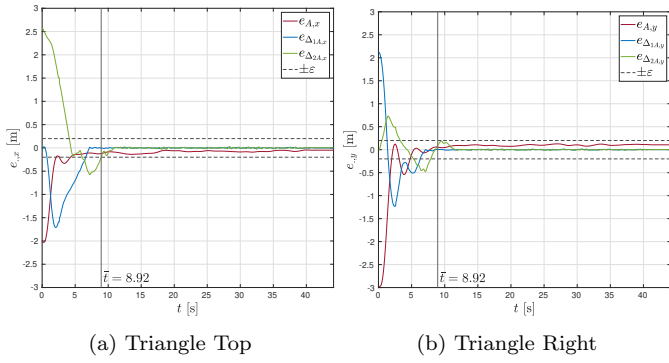


Fig. 9. Formation errors of the team corresponding to the triangle-top shape in the X and Y directions, shown on the left and right, respectively.

which are again inside the unitary circle.

The first experiments have been conducted with the ground free of obstacles. The resulting trajectories are shown in Fig. ???. In the figure, an experiment (the one corresponding to the line segment shape) has been omitted due to limited space. The geometric shape is achieved by the fleet. For the quantitative analysis, the negligible error has been increased to $\varepsilon = 0.20$ m. With this new definition, the formation is achieved after $\bar{t} = 8.92$ s and $\bar{t} = 10.34$ s in the two scenarios respectively. However, the quadrotor is not exactly in the desired position, denoted by the red star. Indeed, let us analyze Fig. ???, which shows the formation errors for the first experiment shown in ??? (considering only this case for the sake of brevity). The ground robots' errors, shown in blue and green, converge to zero, while the drone maintains a non-zero error that is still smaller than ε . The same behavior has been observed in all the tests. The robots G_1 and G_2 , however, keep the desired distances from the drone and from G_2 , respectively. For these reasons, the final geometry of the team (solid line) in Fig. ??? maintains the correct shape and orientation but translated from the desired configuration (dotted black line). Moreover, even if the desired final positions are inverted w.r.t. the initial positions of the ground vehicles, they do not collide. A video collecting the experiments can be found at: <https://youtu.be/wefGtmPAPiA>. This time, the test with the line segment was also included. In this scenario, the geometric shape is achieved at $\bar{t} = 10.04$ s. At the end of all experiments, even the ground vehicles appear

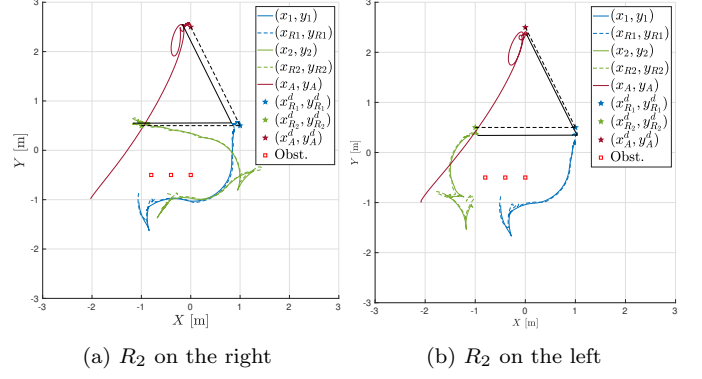


Fig. 10. Trajectories of the three agents with desired and actual interdistances in the experiments with static obstacles.

to oscillate. This can be explained by the small oscillations of the drone around the desired position, which are then mirrored by the unicycles as they maintain the geometric shape. To prove this supposition, a trial was performed using only the ground robots, while the drone's position was mimicked by applying a constant setpoint throughout the test. The plots are not made for the sake of brevity, but the experiment is included in the previous video. In the video, it can be seen that once they reach the desired distance from the setpoint, the two Jetbot halt and hold their position.

Once the validation of formation maintenance and collisions prevention was complete, the next aspect to be investigated was obstacle avoidance, both still and in motion. To begin, static obstacles were considered. Specifically, the first experiment considered the geometric shape and the initial configuration shown in Fig. ???. To be accurate, y_A^d was increased by half a meter; otherwise, the unicycles could not have achieved the formation due to their close proximity to the obstacles. Moreover, this time three cones (in Fig. ???, the red squares) were placed in front of the robots. The resulting trajectories are displayed in Fig. ???. Let us compare it with Fig. ???. In both cases, the geometric shape is formed (in this case $\bar{t} = 15.94$ s) without collisions between G_1 and G_2 . This time, however, the unicycles have to move avoiding the cones on the ground. Consequently, more time is required to reach the formation than in the obstacle-free case. On the one hand, without obstructions, G_1 and G_2 arrive at the desired distance from the drone and G_1 in $\bar{t}_1 = 6.60$ s and $\bar{t}_2 = 8.92$ s, respectively, with:

$$\forall t \geq \bar{t}_1 : |e_{\Delta_{A1,x}}(t)| < \varepsilon, |e_{\Delta_{A1,y}}(t)| < \varepsilon \quad (42)$$

$$\forall t \geq \bar{t}_2 : |e_{\Delta_{12,x}}(t)| < \varepsilon, |e_{\Delta_{12,y}}(t)| < \varepsilon \quad (43)$$

On the other hand, with the three cones, we have $\bar{t}_1 = 9.80$ s and $\bar{t}_2 = 15.94$ s. Moreover, the robot G_2 does not go the shortest route towards the green star, but instead approaches it from the right to avoid collisions with G_1 . To confirm this assumption, another experiment was conducted in which the ground robots' starting positions were swapped, so that the robots began on the same side as their respective desired final positions. Fig. ??? shows the results. The vehicle G_2 this time avoids the obstacles moving on the left-hand side of the cones. This is reflected in the key performance indexes: $\bar{t} = 9.66$ s, $\bar{t}_1 = 8.20$ s, and $\bar{t}_2 = 9.66$ s. A video with the two experiments can be seen at: https://youtu.be/_DwPV7lcr4U.

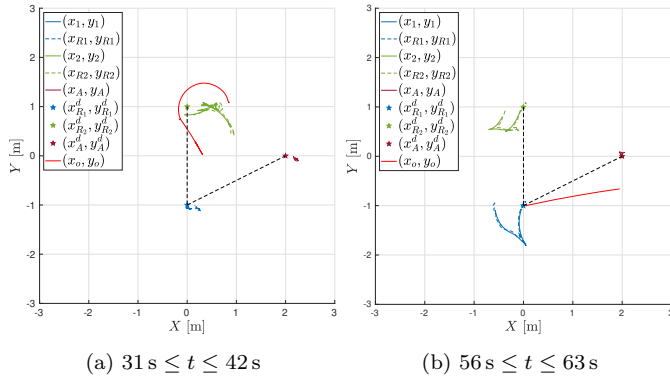


Fig. 11. Trajectories of the three agents with desired and actual interdistances in the experiments with a dynamic obstacle.

In the final test, the cones were removed but a new ground vehicle playing the role of dynamic obstacles was introduced. While the other agents form the desired shape, the robot, guided using a joystick by an operator tries to disrupt the team. The geometric shape and the initial positions of the agents are the ones in Fig. ?? . To better show and analyze the results, the robots' trajectories are plotted not for the entire test, but for some specific, interesting intervals of time. At the beginning of the experiment, the dynamic obstacle is far from the agents, allowing them to reach the formation, as in the scenario free of obstacles in Fig. ?? . Subsequently, however, the jammer moves closer to the Jetbots. When the remote-controlled robot gets close to G_2 , this last moves away, breaking the formation to keep a safe distance from the obstacle. Still when the jammer goes back to being far apart, G_2 will position itself at the desired distance from G_1 . When the obstacle gets close to G_1 , both the two-wheeled robots start moving but for two different reasons. On the one hand, G_1 moves to be at a safe distance from the obstacle. On the other hand, G_2 moves not because it is too close to the obstacle, but to keep the desired distance from G_1 . A video of the test is available at: <https://youtu.be/vXEV-TBfDAY>.

6. CONCLUSION

This paper continues the work by ? and addresses the formation of a heterogeneous fleet composed of two unicycles and a quadcopter. The proposed solution combines feedback linearization, a LMI-based controller, and APFs. The robust formation achievement problem is defined as a linear-quadratic dynamic game, where the optimal control strategy is obtained by solving only one time offline a LMI. To avoid collisions among the ground robots and with any obstacles, a hierarchy is defined between the two-wheeled agents, and APFs are employed. After some simulations, the proposed scheme has been extensively validated through experiments in an indoor arena. The test set included scenarios with static and dynamic obstacles. Results show that in all the cases, the team is able to reach the formation without the crashes of the ground robots. Still there is room for improvement, which opens up to new perspectives. First, to reduce the position tracking error of the drone, one approach is to introduce two additional dynamics: an integral action on the X and an integral

action on the Y . In this way, an analogy can be drawn with a PID controller, similar to the one used with the PD controller. Finally, in this work it was assumed that the Z-controller compensates the gravity, which results in (X, Y) -dynamics independent of the drone's mass. For future work, it would be interesting to include the quadrotor's altitude in the formation dynamics to address changes in the mass, potentially by modelling the overall system as a Linear Parameter-Varying one.

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